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Surname:															
First Name(s):															

Department of Mathematics and Applied Mathematics
 MAM2000W: 2LA (Linear Algebra)

Test 1 Time : 1.5 hours	13 March 2023 Maximum marks: 52 Full marks: 50
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- This question paper consists of 10 pages (including this one).
- Answer all questions in the spaces provided on this question paper. Use the backs of pages when you run out of space.
- Use your *UCT Examination Answer Book* for rough work. We won't take this work in! Only the answers that you write on this question paper will be marked.
- Basic scientific calculators are allowed.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.

DO NOT WRITE BELOW THIS LINE

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	TOTAL
9	8	4	10	3	8	7	3	52

Student Number:

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Question 1. [9 points] Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 4 & 1 & 5 \\ 3 & 0 & 1 \end{pmatrix}$.

(a) Write A^{-1} as a product of elementary matrices. (Gauss reduce and show your working.)

$$\begin{pmatrix} 1 & 0 & 0 \\ 4 & 1 & 5 \\ 3 & 0 & 1 \end{pmatrix} \begin{array}{l} R_2 \rightarrow R_2 - 4R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{array}$$

Elementary matrix corresponding to

$R_2 \rightarrow R_2 - 4R_1$:

$$E_1 = \begin{pmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{pmatrix} R_2 \rightarrow R_2 - 5R_3$$

Elementary matrix corresponding to

$R_2 \rightarrow R_2 - 5R_3$:

$$E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

I_3

Elementary matrix corresponding to

$R_2 \rightarrow R_2 - 5R_3$:

$$E_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{pmatrix}$$

We have $E_3 E_2 E_1 A = I_3$.

Hence $A^{-1} = E_3 E_2 E_1$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(We can also swap E_1 and E_2 .)

Another solution is

$$A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 11 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{pmatrix}$$

Student Number:								
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- (b) Write A as a product of elementary matrices. (Specify the entries of each elementary matrix explicitly.)

$$A = E_1^{-1} E_2^{-1} E_3^{-1}$$

$$= \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{pmatrix}}_{\rightarrow}$$

There are other solutions (depending on what you obtained in 1(b)).

- (c) Suppose B is a square matrix and $B = E_k \dots E_2 E_1 U$ for some elementary matrices $E_k, \dots, E_2, E_1$ and U is in row echelon form. Does this imply that B is invertible? Justify your answer.

No. Take for instance

$$U = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

U is in row echelon form.

For any elementary matrices $E_1, E_2, \dots, E_k$, we get $B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ and B is not invertible.

Question 2. [8 points] Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}$.

- (a) Find all vectors $\mathbf{b}$ for which the equation $A\mathbf{x} = \mathbf{b}$ has:

- (i) infinitely many solutions
- (ii) no solutions
- (iii) a unique solution.

Show your working. [**Hint:** Write $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ and Gauss reduce the associated augmented matrix.]

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & b_1 \\ 1 & 1 & 1 & b_2 \\ 0 & 1 & 2 & b_3 \end{array} \right) R_2 \rightarrow R_2 - R_1$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & b_1 \\ 0 & -1 & -2 & b_2 - b_1 \\ 0 & 1 & 2 & b_3 \end{array} \right) R_3 \rightarrow R_3 + R_2$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & b_1 \\ 0 & -1 & -2 & b_2 - b_1 \\ 0 & 0 & 0 & b_3 + b_2 - b_1 \end{array} \right)$$

(i) infinitely many solutions for all $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ with $b_3 + b_2 - b_1 = 0$.

(ii) no solutions for all $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ with $b_3 + b_2 - b_1 \neq 0$.

(iii) not possible because there is a free variable because there is a column without pivot in the reduced matrix.

(b) Is A singular or non-singular? Give a reason for your answer.

A is singular because according to (a), the equation $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions.

Student Number:								
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Question 3. [4 points] The following is an augmented matrix of a linear system, in the variables stated. Solve the system, giving your answer **in vector form**.

$$\left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 24 \\ 0 & 0 & 0 & 1 & 0 & 8 \\ 0 & 0 & 0 & 0 & 1 & 9 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right) \quad v, w, x, y, z.$$

$$\underline{z = 9}$$

$$\underline{y = 8}$$

$$\underline{x = s}$$

$$\underline{w = t}$$

$$\underline{v = 7 - s - t}$$

Hence

$$\begin{pmatrix} v \\ w \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 7 \\ 0 \\ 0 \\ 8 \\ 9 \end{pmatrix} + s \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad s, t \in \mathbb{R}$$

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Question 4. [10 points] Let

$$A_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 2 & 0 & 0 \\ -7 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix},$$

$$A_4 = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad A_5 = \begin{pmatrix} 4 & 5 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 2 \end{pmatrix}, \quad A_6 = \begin{pmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

In each of (a) to (g) below, a description of a matrix is given. In each case, choose **one** matrix from the list above satisfying the description given. Write **only one of** $A_1, A_2, A_3, A_4, A_5, A_6$ in the appropriate block on the answer sheet. If it is impossible to find a matrix satisfying the description given, write “Impossible”. You may use a matrix more than once, or not at all.

Student Number:

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(a) A matrix that is not row-equivalent to I_3 .

A_5

(b) A symmetric matrix in row echelon form.

A_2

(c) A singular elementary matrix.

Impossible

(d) An upper triangular matrix which is invertible.

A_2 and A_4 are correct

(e) An elementary matrix that is not in row echelon form.

A_1 and A_6 are correct

(f) A matrix which can be written as the product of two elementary matrices, neither of which is an identity matrix.

A_1, A_2 , and A_6 are correct

Also give the two elementary matrices, in the correct order.

A_1 can for instance be written as

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 20 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -17 & 1 \end{pmatrix}$$

A_1

There are many other solutions.

(g) A matrix which is not elementary and not in row echelon form, but can be written as the product of an elementary matrix and a matrix in row echelon form.

A_3 and A_5 are correct.

Give the two matrices in the product, in the correct order.

For instance

$$A_3 = \begin{pmatrix} 1 & 0 & 0 \\ -3.5 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \text{ and } A_5 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix} \begin{pmatrix} 4 & 5 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Student Number:

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Question 5. [3 points] Suppose the $n \times n$ matrix A can be written as $A = E_1 E_2 \dots E_k$, where $E_1, E_2, \dots, E_k$ are elementary matrices. How can we write $2A$ as a product of elementary matrices?

Let F_i be I_n , where the i th diagonal entry is replaced by 2. Then

$$2A = F_1 F_2 \dots F_n E_1 E_2 \dots E_k.$$

Question 6. [8 points] Read the following proof and then answer the questions following it:

Proof: Let the $n \times n$ matrix A be row equivalent to I_n (1)

Then there are elementary matrices $E_1, E_2, \dots, E_k$ such that $E_k \dots E_2 E_1 A = I$ (2)

Hence $A = E_1^{-1} E_2^{-1} \dots E_k^{-1}$, (3)

and therefore A is invertible. From this it follows that A is non-singular. $\square$

(a) State the result proved here.

If a square matrix A is row equivalent to I_n then it is non-singular.

(b) Use the definition of row equivalence to explain how (2) follows from (1).

"A row equivalent to I_n " means that there is a sequence of elementary row operations $e_1, e_2, \dots, e_k$ that turns A to I_n . Hence there are corresponding elementary matrices such that $E_k \dots E_2 E_1 A = I_n$

(c) Why does E_1^{-1} exist?

Student Number:

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E_1^{-1} exists because E_1 is an elementary matrix and we know that every elementary matrix is invertible.

- (d) How does it follow from (3) that A is invertible?

Every matrix E_i^{-1} is invertible, and the product two or more invertible matrices is also invertible.

- (e) Use only the definitions of invertibility and non-singularity to prove that an invertible matrix is non-singular.

Suppose that $A\mathbf{x} = \mathbf{0}$.

Since A is invertible, A^{-1} exists.

We multiply both sides of the equation by A^{-1} on the left.

We get

$$A^{-1}(A\mathbf{x}) = A^{-1}\mathbf{0}.$$

This simplifies to

$$(A^{-1}A)\mathbf{x} = \mathbf{0},$$

and hence

$$\mathbf{x} = \mathbf{0}.$$

Therefore, $\mathbf{x} = \mathbf{0}$ is the only equation for the equation $A\mathbf{x} = \mathbf{0}$, which prove that A is

non-singular

Question 7. [7 points] Consider the following statement:

For every $n \times n$ matrix A , if $A^{2023} = \mathbf{0}$, then $I_n - A$ is invertible.

- (a) Write down the converse of the statement.

For every $n \times n$ matrix A , if $I_n - A$ is invertible then $A^{2023} = \mathbf{0}$.

- (b) Write down the contrapositive of the statement.

For every $n \times n$ matrix A , if $I_n - A$ is not invertible then $A^{2023} \neq \mathbf{0}$.

Student Number:

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(c) Prove or disprove the statement.

The statement is true.

We have

$$(A^{2022} + A^{2021} + \dots + A^2 + A + I_n)(I_n - A) = I_n - \underbrace{A^{2023}}_{=0} = I_n$$

just multiply out!



Hence

$$A^{2022} + A^{2021} + \dots + A^2 + A + I_n$$

is the inverse of

$$I_n - A.$$

(d) Prove or disprove the converse of the statement.

The converse is false.

Take for instance

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

Then

$$I_2 - A = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

is invertible.

But $A^{2023} \neq 0$.

Student Number:

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Question 8. [3 points] Suppose that A , and B are square matrices and $AC = CB$ for some invertible matrix C . If $B^k = I$ for some positive integer k , is it true that $A^k = I$?

The statement is true.

Multiply both sides of the equation $AC = CB$ by C^{-1} on the right.

Then we get $A = CBC^{-1}$,

and therefore,

$$\begin{aligned}
 A^k &= \overbrace{CBC^{-1}CBC^{-1}CBC^{-1} \dots CBC^{-1}}^{k \text{ times } CBC^{-1}} \\
 &= \underbrace{CB^k}_{I} C^{-1} \\
 &= CC^{-1} \\
 &= I
 \end{aligned}$$